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1beta DRAMs are low-powered and have double data rate 5X (LPDDR5X), capable of delivering data at a rate of 8.5Gbps. Multi-pattern lithography, in combination with leading-edge process technology and advanced material capabilities, is required for the manufacture of 1beta node DRAMs. Introduction of enhanced dynamic voltage and frequency scaling extensions core (eDVFS) techniques is used to improve power-saving controls of DRAMs.

1-gamma node DRAMs are coming

This advanced DRAM is expected to unlock exponential growth and opportunities for an increase in the digital economy. It is a major step to secure the semiconductor supply chain. Currently, Micron is a pioneer

in DRAM manufacturing. Recently, it installed extreme ultraviolet (EUV) lithography production equipment at its A3 wafer fabrication unit in Taiwan.

In May 2023, Micron announced that it will be introducing this sophisticated pattern technology (EUV) to manufacture its next-generation DRAM, 1-gamma node. This technology is expected to provide the smallest cell size for DRAM. The downside of EUV lithography is that it needs multi-patterning, allowing 193nm lithography to considerably reduce the size of DRAM. Micron is combining new materials, tools, and novel techniques to improve its multi-patterned alignment.

Challenges faced by DRAM industry

One-die Error Correction Code

(ECC) is a special feature in the DRAM industry that corrects bit errors in DRAM chips, increasing reliability and defect rates. ECC requires additional memory storage where ECC codes are stored at the time of memory write to DRAM. Hamming codes are used in such ECC schemes to provide single-bit and double-bit error detection per burst. Latest DRAMs like LPDDR5 and DDR5 support one-die ECC.

Row hammer. The same or adjacent rows are activated again and again, resulting in loss or change of data contents in the rows that are not addressed. Latest versions of DRAM LPDDR5/DDR5 support refresh management (including DRFM and ARFM) to compensate for this challenge.

Device temperature. Users need to check with the DRAM vendor